

VIRGINIA BEACH BUOY 44014, USA

WMO: 996300

Lat: 36.606N

Lon: 74.840W

Elev: 0

StdP: 14.70

Time Zone: -5.00 (W05)

Period: 94-15

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	28.8	32.0	15.3	12.0	30.5	18.2	13.8	34.0	40.3	38.6	36.9	40.0	18.7	320	0.471

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	3.1	83.2	79.0	82.0	77.7	80.9	76.6	81.0	83.5	79.7	82.4	78.5	81.0	9.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
80.2	156.7	83.0	78.8	149.6	81.9	77.7	143.8	80.8	44.5	83.6	43.2	82.4	42.0	81.4	84.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 33.7	30.1	25.1	25.4	82.9	4.4	1.5	22.2	84.0	19.6	84.9	17.1	85.8	13.9	86.9		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	60.6	45.8	45.3	48.1	53.5	60.4	70.7	77.2	77.4	73.3	65.7	57.0	51.3		
	DBStd	12.70	7.71	6.26	4.87	4.63	4.75	4.84	2.77	2.67	3.27	5.02	6.41	7.07		
	HDD50	525	175	150	91	19	0	0	0	0	0	0	17	73		
	HDD65	2910	595	551	522	346	156	13	0	0	0	53	247	427		
	CDD50	4382	44	19	34	123	323	622	842	850	698	487	228	112		
	CDD65	1292	0	0	0	0	14	184	377	385	248	75	8	1		
	CDH74	6611	0	0	0	0	3	453	2585	2748	782	40	0	0		
	CDH80	439	0	0	0	0	0	8	182	223	26	0	0	0		
(14) Wind	WSAvg	11.6	15.2	14.5	13.0	11.2	9.3	8.5	9.2	9.3	10.3	11.5	12.7	14.1		
(15) Precipitation	PrecAvg	65.4	5.2	5.0	5.4	4.6	4.9	4.9	6.9	6.7	5.9	5.1	5.2	5.5		
	PrecMax	80.7	9.8	10.3	8.5	8.0	9.6	9.0	10.0	12.5	11.5	8.8	8.0	9.1		
	PrecMin	49.9	2.4	2.2	2.5	1.9	2.0	2.4	3.0	3.0	1.5	2.0	0.7	2.5		
	PrecStd	6.9	1.7	1.7	1.5	1.5	1.8	1.6	1.8	2.3	2.4	2.0	1.9	1.8		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	63.9	61.9	61.2	66.1	73.9	81.2	85.0	85.1	82.9	76.9	71.9	68.1		
		MCWB	62.0	59.7	58.1	63.9	71.5	78.4	80.9	80.8	79.0	74.4	69.0	65.9		
	2%	DB	61.4	58.1	58.9	63.9	71.3	79.1	83.1	83.5	80.3	75.3	69.8	65.1		
		MCWB	59.3	55.9	56.5	62.2	69.7	76.2	79.2	79.2	77.1	72.6	67.0	63.7		
	5%	DB	59.5	55.8	56.9	61.9	68.8	77.8	81.9	82.3	78.8	74.1	67.9	62.9		
		MCWB	57.2	52.7	54.6	60.4	67.4	74.9	78.0	77.6	75.6	70.8	65.0	61.1		
	10%	DB	57.0	53.9	55.0	60.0	66.7	76.7	80.9	81.2	77.7	72.7	65.9	61.1		
		MCWB	53.8	50.7	52.8	58.6	65.2	74.0	77.0	76.3	74.5	69.1	63.1	59.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	63.0	61.2	61.2	64.9	73.7	76.9	82.9	82.3	80.3	75.3	68.1	63.9		
		MCDB	63.5	61.5	61.4	65.7	75.0	77.5	84.9	84.7	82.6	76.3	69.1	64.4		
	2%	WB	61.1	58.0	58.7	62.5	71.4	75.8	81.0	81.3	79.3	73.4	65.7	62.3		
		MCDB	61.9	59.2	59.4	63.5	72.8	76.7	83.4	83.9	82.0	74.5	67.6	62.9		
	5%	WB	58.9	54.6	56.9	61.0	70.1	75.0	79.5	80.1	78.0	71.1	64.1	60.2		
		MCDB	60.2	56.2	58.0	61.9	71.5	76.2	81.9	82.8	80.1	73.0	65.9	61.1		
	10%	WB	56.0	51.9	54.5	59.1	68.5	74.1	78.3	78.9	76.5	69.3	62.3	58.4		
		MCDB	58.0	54.1	56.1	60.1	69.8	75.6	80.5	81.6	78.4	71.7	64.1	59.7		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.4	8.3	6.8	5.6	4.7	3.8	3.2	3.1	3.3	4.5	6.1	7.1		
		MCDBR	10.1	11.2	8.8	6.9	5.1	4.0	3.6	3.6	3.6	4.6	7.2	8.7		
		MCWBR	11.1	13.7	9.1	7.1	5.0	4.3	4.6	4.7	4.6	4.4	9.9	11.2		
	5% WB	MCDBR	9.5	12.5	9.0	6.7	5.0	3.0	3.5	3.4	3.6	4.0	8.3	8.7		
		MCWBR	11.6	14.4	10.1	7.0	5.0	3.9	4.9	4.8	4.0	5.0	10.1	10.4		
(40) Clear-Sky Solar Irradiance	taub		0.334	0.342	0.358	0.400	0.445	0.479	0.503	0.495	0.417	0.373	0.341	0.337		
	taud		2.505	2.482	2.464	2.352	2.260	2.216	2.157	2.180	2.391	2.494	2.528	2.520		
	Ebn at Noon		272	284	288	280	268	258	250	250	266	269	267	263		
	Edh at Noon		26	30	33	39	43	45	48	46	35	29	25	24		
(44) All-Sky Solar Radiation	RadAvg		684	952	1325	1725	1955	2103	1991	1807	1487	1166	811	631		
	RadStd		53	93	106	109	146	122	93	127	106	95	67	53		

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46) Station Only	DBAvg	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(47) Regional (33 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
		+0.79	N/A	N/A	N/A	N/A	N/A	N/A	+299	+193

Nomenclature: See separate page